

CEN111 Algorithms and Programming I

Fall 2021-2022

Instructor: Dr. Barış ATA (bata@cu.edu.tr)

Office Hours: Wednesday, 13:00-15:00

Description:

This course aims to teach students the basics of computer programming using C Programming Language. This course focuses on Algorithm design tools: flowcharts, pseudocodes, C programming language primitives, I/O functions, Random Numbers, Operations, Type Conversions, Precedence of Evaluation, Conditions, Loops, Functions, Recursive Functions, Arrays.

Textbooks:

- Problem Solving and Program Design in C —7th ed, Jeri R. Hanly, Elliot B. Koffman
- C How to Program - 8th ed, Paul Deitel, Harvey Deitel

Required Software:

- Raptor Flowchart based Programming Environment
- Microsoft Visual Studio

References:

- C Programming Language (2nd Edition), Brian W. Kernighan, Dennis M. Ritchie
- C Programlama Dili, Rifat Çölkesen
- C Dersi: Programlamaya Giriş, Nergiz Ercil Çağıltay, C. Fügen Elbes, Gül Tokdemir, Çiğdem Turhan

Course Schedule:

Week 1:	Introduction
Week 2:	Algorithmic Thinking Steps
Week 3:	Problem Solving
Week 4:	Representing Information
Week 5:	Typical C Program Development Environment/Structure of a C Program
Week 6:	Variable Names, Data Types, Constants, Declarations
Week 7 :	Operators (Arithmetic, Relational and Logical) and Type Conversions
Week 8 :	Increment and Decrement Operators, Precedence of Evaluations
Week 9 :	Conditional Expressions
Week 10:	Control Flows
Week 11:	Basics of Functions
Week 12:	Recursion
Week 13:	Arrays I
Week 14:	Arrays II

Grading: Midterm 40 %, Final exam 60 %.